Introduction and Math Review Practice 1

Module 1 - Introduction & Math Review · Daily practice sheet · No date

Name _____ Date _____

1a In an experiment to study the chemical decomposition, a drug solution was prepared and a sample was obtained at different time points. The drug concentrations in the samples were as follows.

Time (hr)	Concentration (mg/L)
2	294.3
6	208.1
12	123.8
24	43.8
36	15.5
48	5.5

What is the rate constant for the decrease in concentration?

1b A drug solution decomposes by a first-order process with a rate constant of 0.0866 hr^{-1}. The concentration measured 12 hours after the solution was prepared was 123.8 mg/L. What was the initial concentration of the solution?

1c A drug solution decomposes by a first-order process with a rate constant of 0.0866 hr^{-1}. How much time is required for exactly half the solution to decompose?

1d A drug solution decomposes by a first-order process with a rate constant of 0.0866 hr^{-1}. How much time is required for the original solution to decompose by 90 per cent?

2a In an experiment to study the chemical decomposition, a drug solution was prepared and a sample was obtained at different time points. The drug concentrations in the samples were as follows.

Time (hr)	Concentration (mg/mL)
2	338
6	314
12	278
24	206
36	134
48	62

What is the rate constant for the decrease in concentration?

2b A drug solution decomposes by a zero-order process at 6 (mg/mL)/hr, and the sample taken at 2 hours measured 338 mg/mL. What was the initial starting concentration of the solution?

2c A drug solution decomposes by a zero-order process at 6 (mg/mL)/hr from an initial concentration of 350 mg/mL. How much time is required for exactly half the solution to decompose?

2 (final part, unnumbered)

A drug solution with an initial concentration of 350 mg/mL decomposes by a zero-order process at 6 (mg/mL)/hr. How much time is required for the original solution to decompose by 90%?